STAT210 Statistical Methods I: Linear Model

Lec 11: Multiple Linear Regression - Multicollinearity

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```
library(tidyr) #the pipe (%>%) tool is extremely useful
library(MASS)
library(ggplot2)
library(reshape2)
library(ggcorrplot)
library(car)
```

Section 1

Review

Review: Lecture 01-06

- Lectures 01-06: SLR: least squares estimate, inference (CI, prediction, t-test, F-test), diagnostics, transformations, ANOVA, lack of fit test.
- Lecture 07: Multiple Linear Regression (MLR): least squares estimate.
- Lecture 08: MLR inference (CI, t-test, linear functions, ANOVA).
- Lectures 9-10: Sums of squares, overall and partial F-tests, and ANOVA.
- Lecture 11: multicollinearity

Motivating Example: CDI

```
cdi <- read.csv("data/CDI.csv")
cdi <- cdi[cdi$region==1,]
#correlation heatmap
cdi$physicians_std <- cdi$physicians / cdi$population * 1000
cdi$hospitals_std <- cdi$hospitals / cdi$population * 1000
cdi$crimes_std <- cdi$crimes / cdi$population * 1000
cdi$land_area_std <- cdi$land_area / cdi$population * 1000
cdi_clean <- cdi[, c("land_area_std", "population",
"X18.34", "X65plus", "physicians_std", "hospitals_std",
"crimes_std", "high_school", "bachelor", "poverty",
"unemployment", "capita_income")]
```

Section 2

Linear Dependency in MLR

CDI Example: Scenario 1

- In the CDI example, high_school is the percent of county population 25 years or older who completed 12 or more years of school.
- I am interested a related variable: the percent of county population 25 years or older who completed less than 12 years of school.
- Let's create this new variable and see how it is related to unemployment rate.

CDI Example: Scenario 1

```
cdi_clean_new <- cdi_clean
cdi_clean_new$no_high_school <- 100 - cdi_clean$high_school
summary(lm(unemployment ~ ., data=cdi_clean_new))
```

```
##
## Call:
## lm(formula = unemployment ~ ., data = cdi_clean_new)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1924 -0.9682 -0.1311  0.6427  4.0858
##
## Coefficients: (1 not defined because of singularities)
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.529e+00  5.555e+00  -0.455  0.64999
## land_area_std   2.093e-02  4.719e-02   0.443  0.65852
## population     -4.821e-08  5.534e-07  -0.087  0.93076
## X18.34          1.742e-01  7.373e-02   2.363  0.02027 *
## X65plus         1.676e-01  7.476e-02   2.242  0.02739 *
## physicians_std -2.428e-01  2.280e-01  -1.065  0.28966
## hospitals_std  -3.151e-01  1.714e-01  -1.838  0.06936 .
## crimes_std     -3.499e-03  6.556e-03  -0.534  0.59489
## high_school     1.996e-02  5.017e-02   0.398  0.69164
## bachelor       -1.435e-01  6.570e-02  -2.185  0.03146 *
## poverty         3.179e-01  8.518e-02   3.732  0.00033 ***
## capita_income   1.833e-04  8.293e-05   2.210  0.02962 *
## no_high_school         NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

Linear Dependency in MLR

- Let $x_1, \cdots, x_{p+1}$ denote the predictor variables. Linear dependency means that there exist constants $a_1, \cdots, a_{p+1}$, not all zero, such that

$$a_1 x_1 + a_2 x_2 + \cdots + a_{p+1} x_{p+1} = 0.$$

- Recall that the LSE is given by

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- When there is linear dependency among the predictor variables, the matrix $X^T X$ is singular and cannot be inverted. In this situation, LSE is not unique.

The Design Matrix

- To see this, we will look at a toy example with only two predictor variables that are linearly dependent and n=5

```
X1 = c(1,1, 0, 0)
X2 = c(0,0, 1, 1)
y = rnorm(4)
X = cbind(1, X1, X2)
XTX = t(X) %*% X
XTX
det(XTX)
solve(XTX)
```

The Design Matrix

```
> X1 = c(2,3,4,5,6)
> X2 = c(8,7,6,5,4)
> y = rnorm(5)
> X = cbind(1, X1, X2)
> XTX = t(X) %*% X
> XTX
      X1  X2
   5  20  30
X1 20  90 110
X2 30 110 190
> det(XTX)
[1] -1.110223e-12
> solve(XTX)
Error in solve.default(XTX) :
  system is computationally singular: reciprocal condition number = 5.60719e-19
```

The first consequence of multicollinearity

- In the artificial examples, there is perfect multicollinearity
  - between high_school and no_no_high_school because no_high_school = 100 - high_school.
  - between X1 and X2 because X2 = 10 - X1
- As a consequence, the LSE cannot be computed.
- When there is close to perfect multicollinearity, the matrix $X^T X$ is close to singular. It is numerically unstable to compute the LSEs.

Section 3

Perfectly Uncorrelated / Orthogonal

Orthogonal Predictors

- Consider the following design matrix with two predictor variables that are perfectly uncorrelated. The sample correlation is 0.

```
X1 = c(1,1,1,1,2,2,2,2)
X2 = c(5,6,5,6,5,6,5,6)
y = rnorm(8)
cor(X1, X2)
```

```
## [1] 0
```

Orthogonal Predictors

- y on X1 only

```
anova(lm(y ~ X1))
```

```
## Analysis of Variance Table
##
## Response: y
##           Df Sum Sq Mean Sq F value Pr(>F)
## X1         1 1.3697 1.36975  1.4517 0.2736
## Residuals  6 5.6612 0.94354
```

Orthogonal Predictors

- y on X2 only

```
anova(lm(y ~ X2))
```

```
## Analysis of Variance Table
##
## Response: y
##           Df Sum Sq Mean Sq F value Pr(>F)
## X2         1  1.576 1.57598  1.7334  0.236
## Residuals  6  5.455 0.90917
```

Orthogonal Predictors

- y on X1 and X2. We noticed that

$$SS(\beta_2|\beta_1) = SS(\beta_2)$$

```
anova(lm(y ~ X1 + X2))
```

```
## Analysis of Variance Table
##
## Response: y
##           Df Sum Sq Mean Sq F value Pr(>F)
## X1         1 1.3697 1.36975  1.6765 0.2520
## X2         1 1.5760 1.57598  1.9289 0.2236
## Residuals  5 4.0852 0.81705
```

Orthogonal Predictors

- y on X2 and X1. We noticed that

$$SS(\beta_1|\beta_2) = SS(\beta_1)$$

```
anova(lm(y ~ X2 + X1))
```

```
## Analysis of Variance Table
##
## Response: y
##           Df Sum Sq Mean Sq F value Pr(>F)
## X2         1 1.5760 1.57598  1.9289 0.2236
## X1         1 1.3697 1.36975  1.6765 0.2520
## Residuals  5 4.0852 0.81705
```

Orthogonal Predictors

- Estimated coefficients

```
summary(lm(y ~ X1))$coefficients
```

```
##               Estimate Std. Error   t value  Pr(>|t|)
## (Intercept) -1.5727547  1.0860117 -1.448193 0.1977261
## X1           0.8275714  0.6868541  1.204872 0.2736154
```

```
summary(lm(y ~ X2))$coefficients
```

```
##               Estimate Std. Error   t value  Pr(>|t|)
## (Intercept) -5.2136775  3.7235437 -1.400192 0.2109867
## X2           0.8876873  0.6742276  1.316599 0.2360265
```

```
summary(lm(y ~ X1 + X2))$coefficients
```

```
##               Estimate Std. Error   t value  Pr(>|t|)
## (Intercept) -6.4550346  3.6577561 -1.764753 0.1378765
## X1           0.8275714  0.6391593  1.294781 0.2519598
## X2           0.8876873  0.6391593  1.388836 0.2235676
```

- Are the results expected? Questions?

Section 4

CDI Example: Scenario 2

Motivating Example: CDI

- There are two variables measuring medical resources:
  - physicians_std: number of physicians per 1000 people
  - hospitals_std: number of hospitals per 1000 people

```
summary(lm(unemployment ~ ., data = cdi_clean))$coefficients
```

```
##                     Estimate   Std. Error     t value     Pr(>|t|)
## (Intercept)    -2.529001e+00 5.554713e+00 -0.45528927 0.6499855910
## land_area_std   2.092692e-02 4.719485e-02  0.44341529 0.6585178252
## population     -4.821393e-08 5.533852e-07 -0.08712543 0.9307632387
## X18.34          1.742129e-01 7.373286e-02  2.36275821 0.0202693926
## X65plus         1.676083e-01 7.475521e-02  2.24209461 0.0273852189
## physicians_std -2.428088e-01 2.279720e-01 -1.06508180 0.2896567990
## hospitals_std  -3.150831e-01 1.714495e-01 -1.83775989 0.0693608469
## crimes_std     -3.498817e-03 6.556443e-03 -0.53364557 0.5948878779
## high_school     1.996147e-02 5.016747e-02  0.39789670 0.6916382398
## bachelor       -1.435489e-01 6.569947e-02 -2.18493293 0.0314626822
## poverty         3.179270e-01 8.517992e-02  3.73241776 0.0003299671
## capita_income   1.832641e-04 8.293057e-05  2.20984933 0.0296243877
```

Motivating Example: CDI

- In lecture 10, we used a partial F-test to show that the medical resource variables are useful, although neither is significant after adjusting for all other variables.

```
lm_full <- lm(unemployment ~ ., data= cdi_clean)
lm_reduced <- lm(unemployment ~ . - physicians_std - hospitals_std, data= cdi_clean)
anova(lm_reduced, lm_full)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ (land_area_std + population + X18.34 + X65plus +
##     physicians_std + hospitals_std + crimes_std + high_school +
##     bachelor + poverty + capita_income) - physicians_std - hospitals_std
## Model 2: unemployment ~ land_area_std + population + X18.34 + X65plus +
##     physicians_std + hospitals_std + crimes_std + high_school +
##     bachelor + poverty + capita_income
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1     93 185.97
## 2     91 164.32  2    21.656 5.9965 0.003578 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Motviating Example: CDI

- Let's look at the models with only one of the medical resource variables removed: keep hospitals_std but remove physicians_std

```
cdi_model_reduced <- lm(unemployment ~ . - physicians_std, data= cdi_clean)
summary(cdi_model_reduced)$coefficients
```

```
##                    Estimate   Std. Error    t value     Pr(>|t|)
## (Intercept)   -9.731543e-01 5.363108e+00 -0.1814534 0.8564108625
## land_area_std  2.481018e-02 4.708817e-02  0.5268879 0.5995396198
## population    -1.926484e-07 5.369040e-07 -0.3588135 0.7205567614
## X18.34         1.514694e-01 7.062455e-02  2.1447133 0.0346108397
## X65plus        1.538462e-01 7.368382e-02  2.0879235 0.0395684520
## hospitals_std -4.325524e-01 1.313618e-01 -3.2928317 0.0014082893
## crimes_std    -2.376263e-03 6.475901e-03 -0.3669394 0.7145068381
## high_school    1.904582e-02 5.019673e-02  0.3794235 0.7052477908
## bachelor      -1.590264e-01 6.411902e-02 -2.4801745 0.0149508017
## poverty        3.075974e-01 8.468778e-02  3.6321344 0.0004622367
## capita_income  1.645492e-04 8.110686e-05  2.0287945 0.0453696151
```

Motviating Example: CDI

- Now keep physicians_std but remove hospitals_std
- Medical resource variables are useful; because they are correlated, neither is significant when both are included.

```
summary(lm(unemployment ~ . - hospitals_std, data= cdi_clean))$coefficients
```

```
##                     Estimate   Std. Error    t value    Pr(>|t|)
## (Intercept)    -2.751801e+00 5.624685e+00 -0.4892366 0.625838629
## land_area_std   2.162006e-02 4.779922e-02  0.4523098 0.652110510
## population      1.312893e-07 5.516898e-07  0.2379767 0.812428372
## X18.34          1.772617e-01 7.466053e-02  2.3742359 0.019663309
## X65plus         1.512536e-01 7.517650e-02  2.0119796 0.047146899
## physicians_std -5.123192e-01 1.767818e-01 -2.8980323 0.004693675
## crimes_std     -4.539103e-03 6.615819e-03 -0.6860985 0.494376477
## high_school     1.529577e-02 5.074641e-02  0.3014159 0.763777621
## bachelor       -1.157979e-01 6.476145e-02 -1.7880692 0.077057038
## poverty         2.791540e-01 8.358518e-02  3.3397546 0.001212356
## capita_income   1.753412e-04 8.388166e-05  2.0903398 0.039345649
```

Section 5

Multicollinearity

Multicollinearity

- In practice, predictor variables are often neither perfectly correlated nor perfectly uncorrelated.
- In practice, predictor variables are often correlated, sometimes highly correlated. This is called multicollinearity.

Consequences of Multicollinearity

- Multicollineary does not affect the overall fit of the model or the prediction, as long as there is no extrapolation.
- The variance of the LSEs are inflated. Recall that $Var(\hat{\beta}) = \sigma^2(X^TX)^{-1}$. $(X^TX)^{-1}$ can have large diagonal elements when it is close to singular (the columns of X are nearly dependent).

Consequences of Multicollinearity

- When X_i and X_j are highly correlated
  - they provide similar information about the response variable.
  - the contribution of x_i given x_j is already in the model is small.
- Similar issues when a predictor is highly correlated with a group of other predictors.

Consequences of Multicollinearity

- When there is multicollinearity, it is difficult to separate their individual effects on the response variable.
- In MLR, we interpret the coefficient of a predictor variable as the effect of that variable on the response when holding all other predictors constant. This is not practically possible when the predictors are highly correlated.

How to Deal with Multicollinearity?

- For prediction purpose, multicollinearity is not a big issue as long as there is no extrapolation.
- For inference purpose, multicollineary is an issue.
- How to detect it?
  - correlation matrix
  - variance inflation factor (VIF)

Section 6

Detection

Motivating Example: mtcars

- cars dataset:
  - y: mpg (miles per gallon)
  - x1: disp (displacement, cu.in.)
  - x2: hp (horsepower)
  - x3: wt (weight, 1000 lbs)

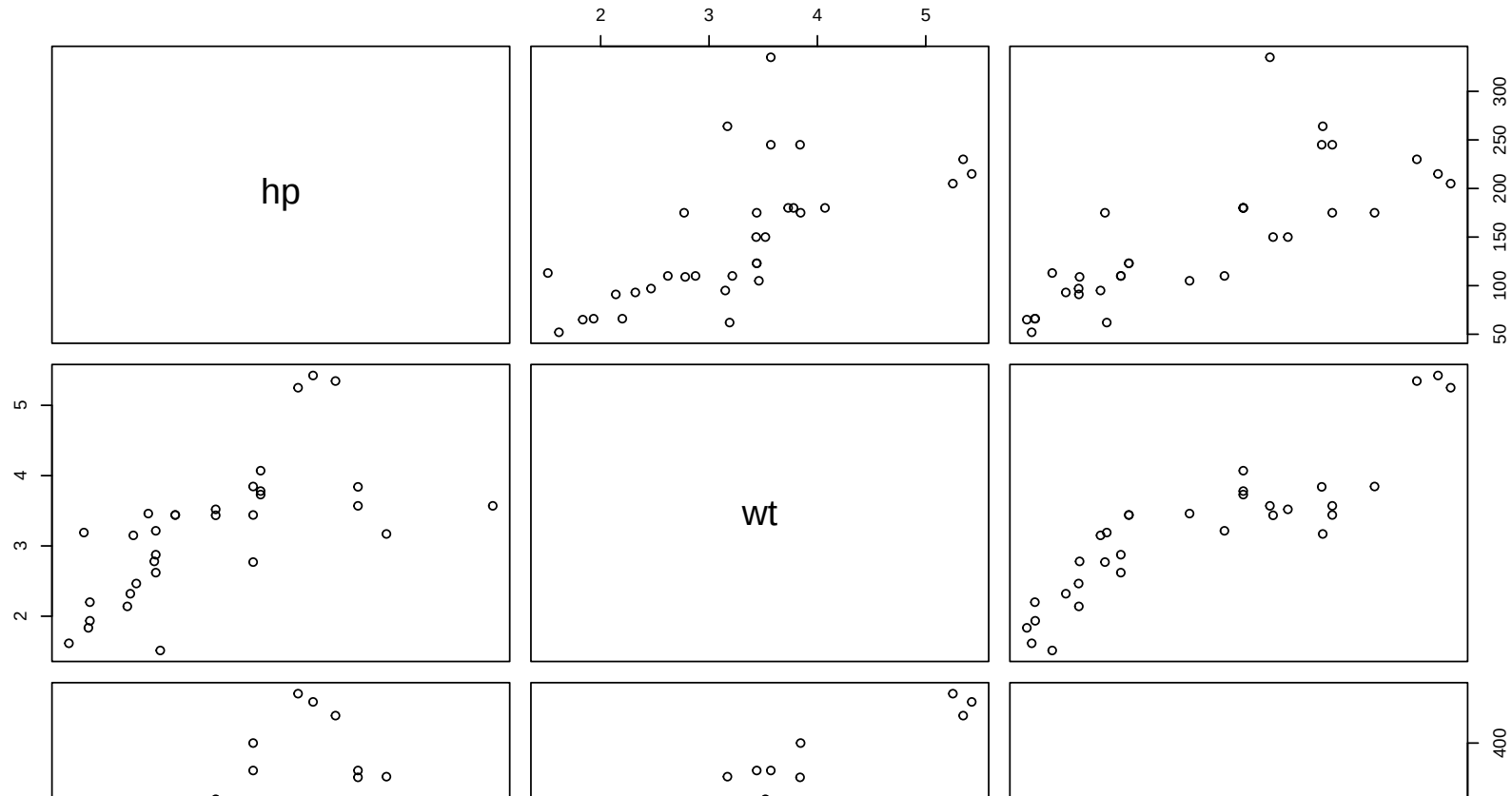
```
cor(mtcars[,c("hp", "wt", "disp")])
```

```
##             hp        wt      disp
## hp   1.0000000 0.6587479 0.7909486
## wt   0.6587479 1.0000000 0.8879799
## disp 0.7909486 0.8879799 1.0000000
```

Motivating Example: mtcars

- Pairwise scatterplots also show that `disp` is highly correlated with `hp` and `wt`.

```
pairs(mtcars[,c("hp", "wt", "disp")])
```



Motivating Example: mtcars

- Based on the correlation matrix, `disp` is highly correlated with `hp` and `wt`.
- We can qualify the correlation between `disp` and (hp, wt) using the multiple R-squared from regressing `disp` on `hp` and `wt`, which is the proportion of variation in `disp` explained by `hp` and `wt`.

```
lm_disp <- lm(disp ~ hp + wt, data=mtcars)
summary(lm_disp)$r.squared
```

```
## [1] 0.8634722
```

Section 7

VIF

Definition of Variance Inflation Factor (VIF)

- The variance inflation factor (VIF) for a predictor variable X_j is defined as

$$VIF_j = \frac{1}{1 - R_j^2},$$

  where R_j^2 is the multiple R-squared from regressing X_j on all other predictor variables.
- In the mtcars example, R_{disp}^2=0.863, so

$$VIF_{disp} = \frac{1}{1 - 0.863} = 7.30.$$

- The `vif` function in the `car` package can compute VIFs for all predictor variables in a linear model.

Example: mtcars

```
vif(lm(mpg ~ disp + hp + wt, data=mtcars))
```

```
##     disp       hp       wt
## 7.324517 2.736633 4.844618
```

Interpretation of VIF

- When there is no multicollinearity, R_j^2 is close to 0 and VIF_j is close to 1.
- When there is multicollinearity, R_j^2 is close to 1 and VIF_j is large.
- A common rule of thumb is that if $VIF_j > 5$ or 10, then there is serious multicollinearity involving X_j.

What to do about Multicollinearity?

- One intuitive method is to remove some of the correlated predictor variables from the model.
- Another method is to combine the correlated predictor variables into a single predictor variable, e.g., by taking their average.
- More advanced methods include principal component regression and ridge regression, which we will cover in future courses.

mtcars Example

- Does removing `disp` help with multicollinearity?

```
vif(lm(mpg ~ hp + wt, data=mtcars))
```

```
##       hp       wt
## 1.766625 1.766625
```

mtcars Example

- Also compare the two models to check whether removing `disp` hurts the model fit.

```
lm_full <- lm(mpg ~ disp + hp + wt, data=mtcars)
lm_reduced <- lm(mpg ~ hp + wt, data=mtcars)
anova(lm_reduced, lm_full)
```

```
## Analysis of Variance Table
##
## Model 1: mpg ~ hp + wt
## Model 2: mpg ~ disp + hp + wt
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     29 195.05
## 2     28 194.99  1   0.05708 0.0082 0.9285
```

```
#alternative, we can use anova(lm(mpg ~ hp + wt + disp, data=mtcars))
```

- So removing disp does not hurt the model fit significantly, and it helps with multicollinearity.